# KARTHIKEYAN V

Tiruchengode, Tamil Nadu, India | +91 9345568688 | vkarthikeyan020@gmail.com
linkedin.com/in/yourprofile | github.com/yourprofile | hackerrank.com

## Professional Summary

Electronics and Communication Engineering graduate with dual expertise in software development (Python, C, SQL) and practical hardware diagnostics. Founder and lead hardware technician of an independent electronics service center, experienced in component-level troubleshooting, motherboard diagnostics, and embedded systems integration. Seeking roles in Embedded Systems, Hardware Support Engineering, or Software Development.

## Technical Skills

- **Programming Languages:** Python, C, Java (Basics), SQL, JavaScript
- **Hardware & Diagnostics:** Component-level troubleshooting, SMD rework & soldering, multimeter/oscilloscope testing, display circuit debugging, power IC fault isolation, BIOS/firmware flashing
- **Embedded & Tools:** Microcontrollers (Arduino, STM/ESP basics), PCB design basics, Git.
- **Web & Frameworks:** HTML5, CSS3, OpenCV, basic REST APIs

## Work Experience

**Founder & Lead Hardware Diagnostic Technician** *2024 – Present*
*Mobile & Laptop Service Center*, Tiruchengode, Tamil Nadu

- Operated an independent mobile and laptop repair center, diagnosing complex hardware faults and software issues across 500+ consumer devices.
- Executed component-level PCB troubleshooting and microsoldering for motherboard faults, short circuits, and damaged power rails using schematics and multimeters.
- Performed chip-level BIOS reflashing, firmware recovery, and low-level data retrieval on unresponsive systems.
- Handled end-to-end technical operations, inventory management for genuine electronic components, and client diagnostics.

**Python Developer Intern** *June 2023 – July 2023*
*Gateway Software Solutions*, Coimbatore, Tamil Nadu

- Developed automation scripts in Python applying object-oriented programming to process and sanitize datasets.
- Designed backend routines and structured SQL database queries to execute reliable CRUD operations.
- Maintained version control workflows using Git and participated in technical review sessions.

## Technical Projects

**Real-Time Computer Vision Fire Detection System** | *Python, OpenCV*

- Engineered an image-processing pipeline capable of early-stage fire and smoke detection from real-time video feeds.
- Applied color segmentation, contour filtering, and spatial thresholding to reduce false positives in dynamic lighting.
- Optimized frame processing rates to ensure low-latency warning triggers for surveillance systems.

**Hardware & Signal Interfacing Implementations** | *Embedded C, Microcontrollers*

- Configured microcontroller firmware to decode and convert multi-channel radio control signals (PWM to PPM stream) with minimal jitter.
- Built diagnostic test benches to trace and isolate ribbon cable signal degradation from display driver IC failures.

## Education

**Hindusthan College of Engineering and Technology**, Coimbatore, Tamil Nadu
*Bachelor of Engineering in Electronics and Communication Engineering* *2020 – 2024*
CGPA: **7.8 / 10.0**

## Achievements & Certifications

- **Mapathon 2023:** Participant in the national geospatial mapping initiative organized by IIT Bombay, ISRO, and AICTE.
- **Learnathon (2021, 2023):** Completed hands-on technical tracks focused on software problem-solving.
- **HackerRank Profile:** Regular problem solving on data structures and core algorithms (`vkarthikeyan020`).